Exercise 4: AST Patterns

1 Regular Expressions as Terms

Use the following alphabet for regular expressions:

$$\begin{aligned}\mathcal{V} &= \{x, y\},\\ \mathcal{F}^0 &= \{a, b, c\},\\ \mathcal{F}^1 &= \{\mathit{star}\},\\ \mathcal{F}^2 &= \{\mathit{concat}, \mathit{alt}\}.\end{aligned}$$

All other $\mathcal{F}^n$ are empty. The signature is $\mathcal{F} = \{a, b, c, \mathit{star}, \mathit{concat}, \mathit{alt}\}$.

The constants a, b, and c represent individual characters. $\mathit{concat}(R, S)$ represents concatenation RS, $\mathit{alt}(R, S)$ represents alternation $R \mid S$, and $\mathit{star}(R)$ represents the Kleene star R^*. Here, R and S stand for arbitrary terms. Associate concatenation to the left.

a. (i) Write the regular expression $((a \mid b) \mid (a \mid b))$ as a term.
(ii) Write a rewrite step to check for redundant alternation $(R \mid R)$, for an arbitrary R, and replace it with R only. Give the set of all variables used in the rules.

b. Given the regular expression $((a \mid b)^*)^*$ as a term:

$$\mathit{star}(\mathit{star}(\mathit{alt}(a, b)))$$

Write a rewrite step (l, r) to replace duplicate Kleene stars R^{**} with a single one R^*, and give the set of all variables used in the rewrite rules.

c. (i) Write the regular expression $((abc) \mid (abc)^*)$ as a term.
(ii) Write a rewrite step (l, r) to replace covering alternations $(R \mid R^*)$ with the covering part (only R^*), and give the set of all variables used in the rewrite rules.

2 Instances and Substitutions

Use the same function symbols as in task 1, with the set of variables $\mathcal{V} = \{x, y\}$ and the set of constants $\mathcal{F}^0 = \{a, b, c\}$.

a. Consider the terms

$$t = \mathit{alt}(\mathit{concat}(x, y), \mathit{star}(x)),$$
$$s = \mathit{alt}(\mathit{concat}(\mathit{alt}(a, b), c), \mathit{star}(\mathit{alt}(a, b))).$$

(i) For each of the two terms, state whether it is a ground term or not. Explain your answer.
(ii) Provide a valid substitution σ for term t so that $\sigma(t) = s$.
(iii) Is $t \gtrsim s$ or $s \gtrsim t$? Explain your answer.

b. Consider the terms

$$t = \mathit{concat}(x, \mathit{star}(y)),$$
$$s = \mathit{concat}(\mathit{alt}(y, a), \mathit{star}(x)).$$

(i) For each of the two terms, state whether it is a ground term or not. Explain your answer.
(ii) Provide a valid substitution σ for term t so that $\sigma(t) = s$.
(iii) Is $t \gtrsim s$ or $s \gtrsim t$? Explain your answer.